

Injection Moulding - Screw Tip & Non-Return Valve (Check Ring) Assembly

1. Overview

On a reciprocating-screw injection moulding machine the screw-tip assembly lets the screw act as a plunger during injection. It comprises the screw tip (nose cone), the non-return valve (NRV) - either a sliding check ring or a ball check - its seat, and a retainer.

2. Non-return valve (NRV) function [nrv]

During recovery (screw rotation) molten polymer flows forward past the open check ring into the shot chamber ahead of the screw. During injection the ring slides back and seats against the seat face, blocking backflow, so the screw delivers a repeatable shot and holds a stable cushion. The NRV is what makes shot weight and cushion repeatable.

3. Wear and clearance

As the ring, seat and sealing faces wear, the valve no longer seals fully. Melt leaks back past the ring during injection, so shot weight drifts and the cushion becomes unstable or falls away. Radial or face clearance beyond the OEM limit indicates replacement. Abrasive (glass-filled) and corrosive (PVC, flame-retardant) resins accelerate this wear markedly.

4. Inspection

Remove the assembly; measure ring OD, seat ID and face wear; check for scoring, peening and ring hang-up. Compare against the OEM tolerance class. A worn or sticking ring is the most common mechanical cause of an unstable cushion and inconsistent suck-back.

5. Replacement and selection

Choose a free-flow (sliding-ring) valve for fast recovery, or a locking / positive shut-off valve for high-precision and shear-sensitive resins. Torque the assembly to spec and verify a repeatable cushion over at least 20 shots before releasing the machine.